Experiment No. 11

Routing at the Network Layer Using Static Routing and RIP in Cisco Packet Tracer

Aim

To implement routing at the network layer using Cisco Packet Tracer by configuring static routing and the Routing Information Protocol (RIP). To establish communication between computers connected to different networks, configure router interfaces and routing tables, and verify packet transmission using the ping command and Simulation Mode.

Procedure

Part A: Static Routing in Cisco Packet Tracer

1. Open Cisco Packet Tracer and create a new network topology. Add two routers, two switches, and four PCs to the workspace. Arrange the devices so that each router connects to a local network and both routers are connected to each other.
2. Connect PC0 and PC1 to Switch0 using Copper Straight-Through cables. Connect Switch0 to Router0. Similarly, connect PC2 and PC3 to Switch1 and connect Switch1 to Router1. Establish a connection between Router0 and Router1 using their appropriate Ethernet interfaces.
3. Configure the IP addresses of the computers using **Desktop → IP Configuration**. Use the following addressing scheme:
    - PC0: 192.168.1.2, subnet mask 255.255.255.0, gateway 192.168.1.1.
    - PC1: 192.168.1.3, subnet mask 255.255.255.0, gateway 192.168.1.1.
    - PC2: 192.168.2.2, subnet mask 255.255.255.0, gateway 192.168.2.1.
    - PC3: 192.168.2.3, subnet mask 255.255.255.0, gateway 192.168.2.1.
4. Configure the LAN interfaces of the routers. Assign 192.168.1.1/24 to Router0's LAN interface and 192.168.2.1/24 to Router1's LAN interface. Enable both interfaces using the no shutdown command.
5. Configure a separate IP network for the link between the routers. For example, assign 10.0.0.1/30 to Router0's inter-router interface and 10.0.0.2/30 to Router1's corresponding interface. Make sure that the interfaces connected to each other are correctly identified and enabled.
6. Open Router0's CLI and enter the following commands to configure a static route to the remote LAN:

7. enable
8. configure terminal
9. ip route 192.168.2.0 255.255.255.0 10.0.0.2

end

10. Open Router1's CLI and configure a route to Router0's LAN:
11. enable
12. configure terminal
13. ip route 192.168.1.0 255.255.255.0 10.0.0.1

end

14. Verify the routing configuration by entering show ip route on each router. The routing table should contain the directly connected networks and the static route to the remote LAN. If the route is missing, check the next-hop address and interface configuration.
15. Test communication between PCs within the same LAN using the ping command. Then open PC0's Command Prompt and execute ping 192.168.2.2 to test communication with PC2 on the remote LAN. Repeat the test in the opposite direction.
16. Switch to Simulation Mode and send a Simple PDU from a PC in one network to a PC in the other network. Observe the packet as it travels through the source switch, source router, inter-router link, destination router, and destination switch. This demonstrates how routers use routing tables to forward packets between different networks.

Part B: RIP Routing in Cisco Packet Tracer

11. Use a separate topology or modify the existing topology to implement dynamic routing. Keep the two routers connected to their respective LANs and ensure that all interfaces have valid IP addresses.
12. Open Router0's CLI and configure RIP Version 2:

enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.1.0

network 10.0.0.0

end

13. Open Router1's CLI and configure RIP Version 2 for its connected networks:

enable

configure terminal

router rip

version 2

no auto-summary

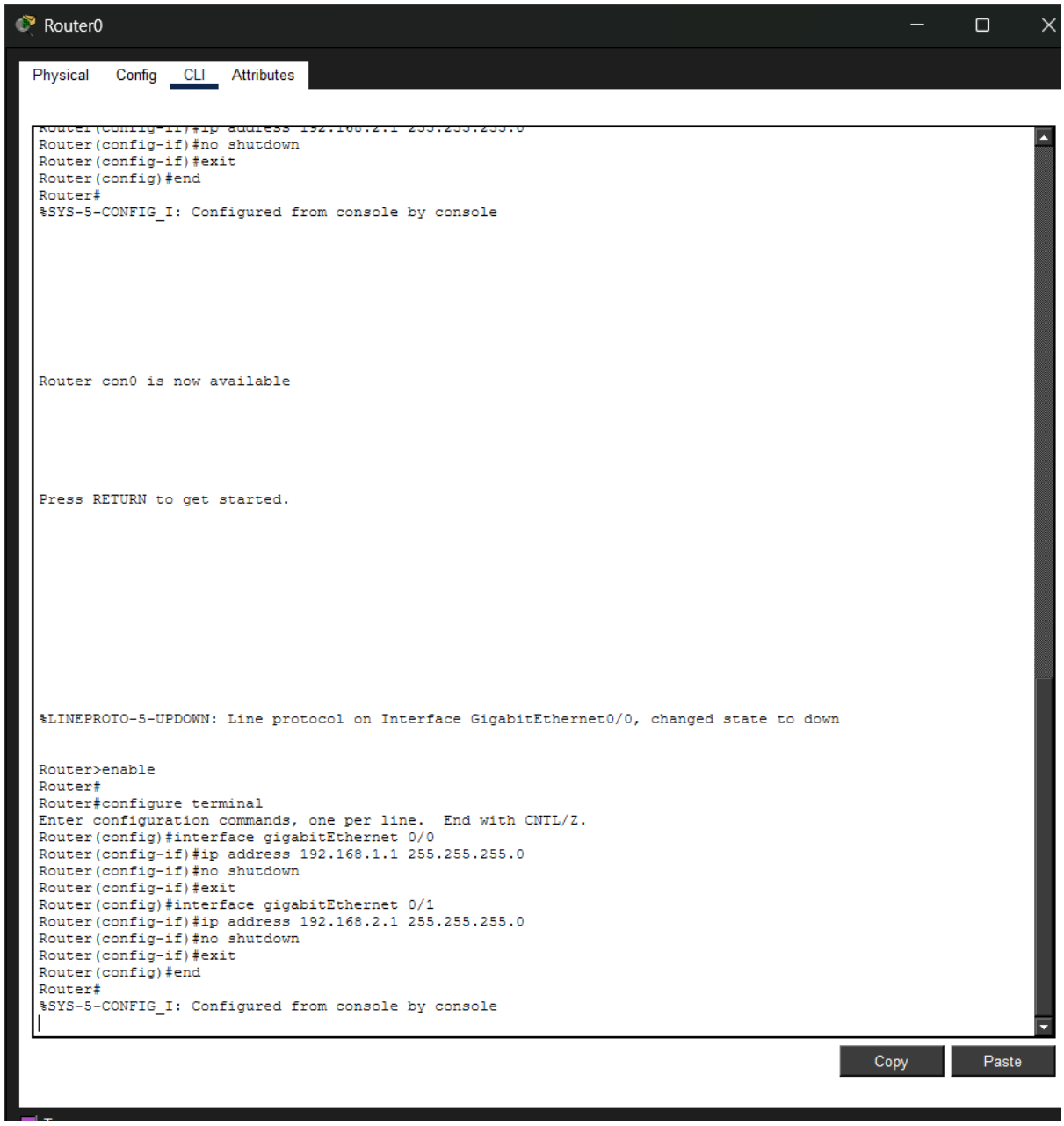
network 192.168.2.0

network 10.0.0.0

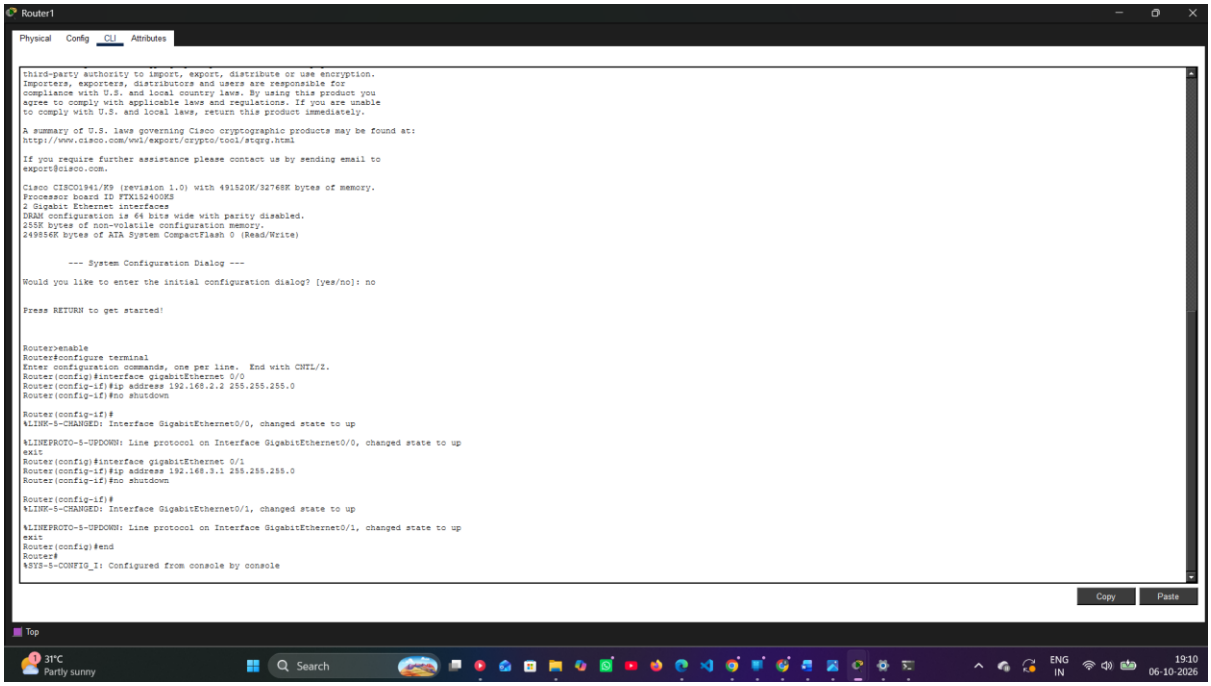
end

14. Allow time for the routers to exchange routing information. RIP enables routers to learn routes to remote networks dynamically instead of requiring the administrator to enter every remote route manually.
15. Execute show ip route on both routers. Check for routes learned through RIP, commonly marked with the letter R. Verify that each router has learned how to reach the LAN connected to the other router.
16. Test communication between PC0 and PC2 using ping 192.168.2.2. Also test communication from PC2 to PC0 using ping 192.168.1.2. If the ping fails, verify the IP addresses, subnet masks, default gateways, router interfaces, and RIP network statements.
17. Use Simulation Mode to observe packet transmission across the network. Compare the static routing and RIP configurations. In static routing, routes are manually configured, whereas RIP allows routers to exchange routing information and learn remote networks automatically.

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         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#no shutdown

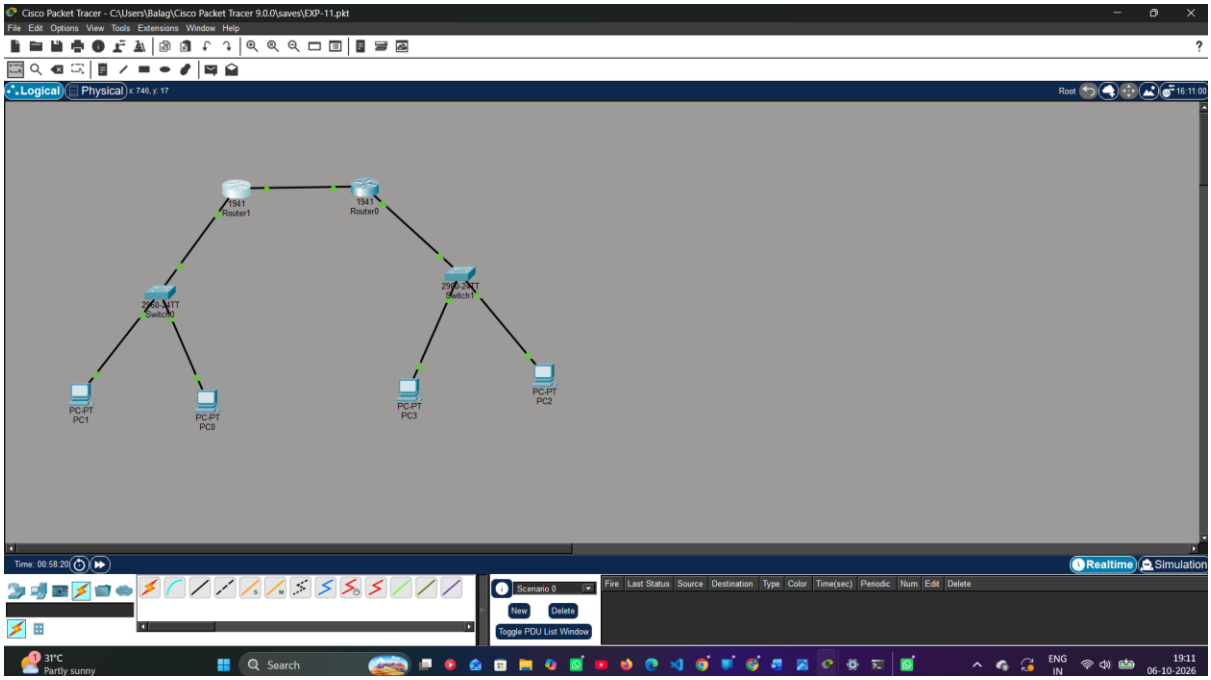
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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```
Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Result

Thus, routing at the network layer was studied and implemented using Cisco Packet Tracer. Static routes were configured manually to enable communication between separate networks. RIP Version 2 was also configured to allow routers to exchange routing information dynamically. Routing tables were verified using show ip route, and end-to-end connectivity was tested using the ping command and Simulation Mode.